

ROWLEY ISLAND, NU, Canada

WMO: 710880

Lat: 69.067N

Lon: 79.065W

Elev: 135

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.1	-32.6	N/A	N/A	N/A	N/A	N/A	N/A	35.0	-8.7	31.3	-12.5	10.2	320	1.141

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	56.7	N/A	53.7	N/A	51.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.2	29.8	26.6	DB	-38.2	62.0	3.6	3.6	-40.8	64.6	-42.9	66.7	-44.9	68.7	-47.5	71.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-17.6	-18.1	-11.8	2.1	20.2	34.0	44.6	41.8	32.5	21.5	4.3	-8.8
	DBStd	23.75	9.84	9.90	10.27	9.31	7.84	4.92	4.34	4.86	4.54	7.43	11.19	10.80
	HDD50	13796	2096	1908	1914	1436	925	480	176	258	524	885	1370	1824
	HDD65	19261	2561	2328	2379	1886	1390	930	634	720	974	1350	1820	2289
	CDD50	10	0	0	0	0	0	0	7	3	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.3	11.8	11.9	12.2	13.0	13.1	13.0	12.6	13.3	14.9	16.0	14.9	13.2
Precipitation	PrecAvg	9.5	0.4	0.3	0.4	0.5	0.6	0.7	1.2	1.8	1.3	1.2	0.7	0.5
	PrecMax	17.0	1.2	1.0	1.1	2.3	1.3	1.6	3.9	2.7	3.1	2.6	2.5	1.2
	PrecMin	4.9	0.0	0.0	0.0	0.1	0.1	0.3	0.4	0.4	0.2	0.3	0.1	0.2
	PrecStd	2.9	0.3	0.2	0.3	0.5	0.3	0.4	0.7	0.6	0.6	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	17.1	20.3	31.3	36.7	53.7	61.6	60.1	46.8	34.6	29.6	25.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	9.1	7.3	12.4	25.2	33.6	47.7	58.0	55.6	42.8	33.0	27.1	18.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	3.1	1.3	8.5	20.5	32.3	44.1	55.2	52.5	40.5	32.0	23.9	12.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	-3.2	-3.6	3.9	16.2	30.8	40.9	52.8	49.7	38.6	30.8	20.4	7.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.9	11.8	12.7	8.9	6.5	11.1	9.6	5.4	6.6	10.1	10.5
		MCDBR	15.5	14.9	15.4	13.5	8.9	12.0	15.5	14.5	8.4	5.3	11.2	15.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.162	0.198	0.244	0.315	0.322	0.320	0.342	0.324	0.292	0.235	0.157	N/A
	taud		1.844	2.031	2.142	1.965	2.089	2.243	2.357	2.506	2.520	2.201	1.923	N/A
	Ebn at Noon		17	188	251	255	271	277	266	260	239	168	17	0
	Edh at Noon		2	14	24	41	41	36	31	24	18	11	1	0
All-Sky Solar Radiation	RadAvg		6	146	673	1420	1942	2106	1765	1160	574	159	19	0
	RadStd		1	13	22	48	89	119	135	76	59	20	3	0

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	+1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page